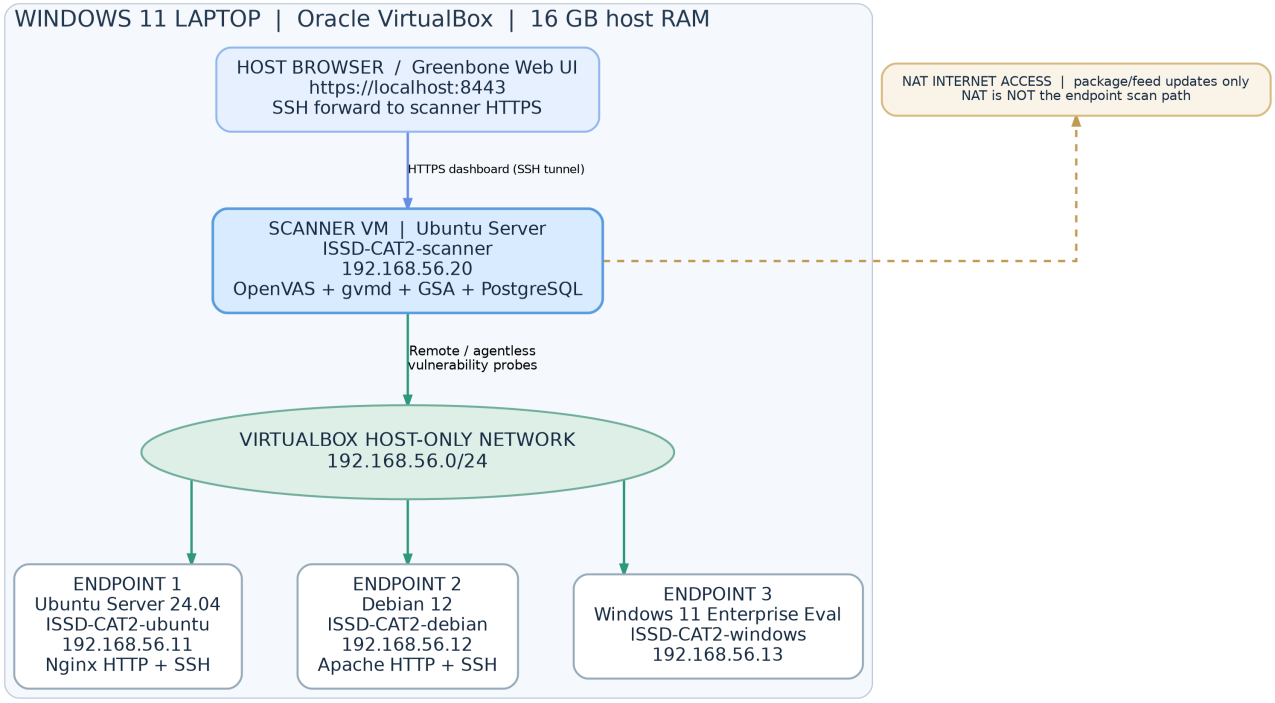
Task 1 | Network-Based Vulnerability Assessment

Evidence completion and lecturer demonstration guide | 17 marks | Members 1-3

Implementation verified: Oracle VirtualBox hosts a Greenbone Community Edition/OpenVAS scanner and three target VMs. All three vulnerability scans completed and the unchanged PDF exports are preserved in the companion ZIP.

Topology



The remote OpenVAS scanner (.20) connects to Ubuntu (.11), Debian (.12) and Windows (.13) through 192.168.56.0/24 host-only networking. The separate NAT interface is for updates. Greenbone is operated from the laptop browser over an SSH tunnel.

Verified findings

Endpoint	Critical	High	Medium	Low	Source report
Ubuntu .11	1	0	0	2	M3_Ubuntu_OpenVAS_Report.pdf
Debian .12	1	0	0	2	M3_Debian_OpenVAS_Report.pdf
Windows .13	0	0	0	1	M3_Windows_OpenVAS_Report.pdf
Total	**2**	**0**	**0**	**5**	**7 displayed findings**

Findings are specific to each PDF export: **QoD at least 70**, excluding Log, Debug and False Positive. The reports display Ubuntu 3/19, Debian 3/17 and Windows 1/3 results after filtering. Mixed-host dashboard overviews were archived rather than used as endpoint-level totals.

Main observations and responses

Ubuntu and Debian - Critical, CVSS 9.8: Both test Linux endpoints accepted the default Vagrant SSH account credentials. This is verified access to an account, not proof that every CVE linked in Greenbone's generic default-credentials reference applies. Mitigation: unique secure credentials or appropriately configured SSH keys, with restricted SSH access; avoid breaking Vagrant access.

Ubuntu and Debian - Low: Each showed a TCP timestamp exposure (CVSS 2.6) and ICMP timestamp reply exposure (CVSS 2.1). Review exposure, firewalling and protocol settings according to the lab requirements.

Windows - Low: TCP timestamp information disclosure (CVSS 2.6). A low finding count does not prove the OS is free of vulnerabilities; unauthenticated scans and firewall rules limit visibility.

What the tool proves

OpenVAS, operated through Greenbone, completed remote agentless vulnerability scans against three different operating systems. Greenbone Management (gvmd) stored and correlated scan results and its web UI exported reports. The original Task 1 phrase about endpoint-specific agents is ambiguous for a network-based assessment and still needs lecturer clarification.

Outstanding evidence and demo checklist

1 Member 1 - Ubuntu OS/IP proof

Capture a genuine screenshot showing hostname, Ubuntu OS version and host-only 192.168.56.11. The provided M4-02 screenshot does not prove this.

2 Member 1 - four-VM overview (optional)

Capture VirtualBox Manager including scanner plus all three endpoints. The copied M4-01 screenshot shows only three endpoint VMs.

3 Member 2 - final storage

Capture df -h /, verifying the expanded scanner root (~140 GB). The earlier 61 GB screenshot is in the archive because it does not show final capacity.

4 Member 2 - healthy Greenbone stack

Capture docker compose ps for scanner services after restarting the stack. Setup and feed screenshots already exist.

5 Member 3 - explain results and limits

Use the corrected source-PDF extracts in Member3 and original PDFs. Describe SSH default credentials, low timestamp findings, filtering, and lack of Windows authentication.

6 Resolve Task 1 agent wording

Ask the lecturer whether endpoint agents are compulsory for Task 1, even though the selected OpenVAS workflow is agentless. Do not claim unverified agent collection.

Short live demonstration order

1. Show network diagram and VM/IP identity. 2. Show three scanner-to-host tests. 3. Show Greenbone targets, scan tasks and feeds. 4. Open Ubuntu, Debian and Windows completed reports. 5. Explain critical and low findings and mitigations. 6. Acknowledge filtering and agent requirement ambiguity.

Data handling: Original scan PDFs may reveal working credentials and service information. Retain the ZIP privately; do not publish it to the public GitHub repository.